

Public-binding interfaces for EndRing PKE

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Result

This batch encoded the requested **propose -> evaluate -> propose if useful -> evaluate** methodology at the public-binding layer of an EndRing-based PKE search. Two mutually blind producers generated materially different candidate relations. Their bytes were frozen before a validator, red team, and T5 reviewer independently evaluated the five prespecified joints. Both raw, candidate-specific independence checks passed. [Method] [Snapshot] [Initial]

The result is **two typed obstructions, zero surviving public-binding interfaces, and zero PKE candidates**. Candidate A, RCF-Bind, has four broken joints and one inconclusive joint. Candidate B, BDEB, also has four broken joints and one inconclusive joint. No single repair could change either disposition, so the optional revision slot was skipped under the frozen early- stop rule. [Final] [Decision]

No secure or novel PKE, IND-CPA/IND-CCA theorem, complete reduction to the supersingular endomorphism-ring problem, implementation, parameter set, benchmark, or impossibility theorem was established. The round was analytical and zero-run: no scientific, implementation, formalizer, experiment, benchmark, or parameter-selection run occurred. Candidate-specific failures do not imply that EndRing PKE is impossible. [Final]

Methodology encoded in the repository

The batch begins before a PKE wrapper. A candidate must define a public, polynomial relation

$R_E(x, y)$ in $\{0,1\}$

on a bare supersingular curve E drawn from the exact declared EndRing challenge distribution. **Sample** and **Eval** may use only public information. For every honest y , every accepted preimage must lie in one public equivalence class, and that equivalence must preserve the functionality used for actual-message recovery. One uniform $\text{Inv}(E, B, y)$ must work for **every** valid complete evaluable basis B of $\text{End}(E)$, including arbitrary valid $\text{GL}_4(\mathbb{Z})$ presentation changes. [Method] [Schema]

The security direction is frozen as

```
scheme attacker
-> accepted preimage or functionality-preserving class
-> complete evaluable basis of  $\text{End}(E)$ .
```

The reverse implication—an EndRing solver can decrypt or invert—is an attack route, not a security reduction. Interface one-wayness is not automatically IND-CPA. Complete-ring necessity needs either a non-factorization result over the declared smaller-witness class or a theorem reconstructing a saturated, evaluable rank-four ring from the smaller witness. [Method]

Two blind proposal tasks were assigned different central relation families. Candidate A could not use a walk endpoint, selector, parity mask, or finite point-evaluation tuple. Candidate B had to differ in domains, relation, equivalence, public operations, and inverse. After the immutable snapshot, T1–T2, T3–T4, and T5 had separate owners. Composition copied owner verdicts without voting. A third proposal was permitted only if one smallest repair could plausibly change a disposition. [Plan] [Snapshot]

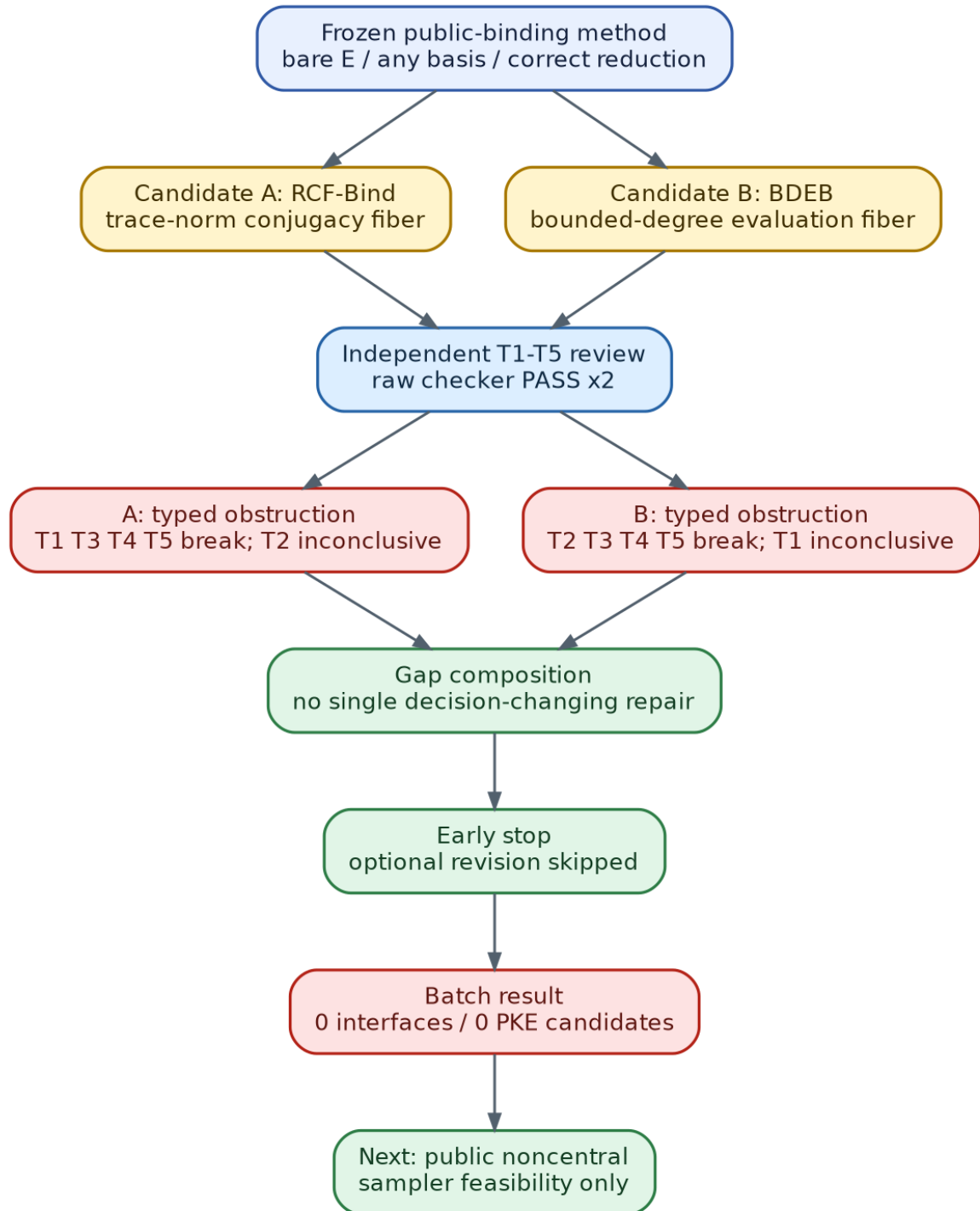


Figure 1: Candidate and review flow. Yellow nodes are proposals, blue is independent review, red is a scoped rejected interface, and green is the controlled stop and future action. The diagram is navigation, not evidence.

Joint	Frozen security/interface question
T1	Bare-E public generation and fixed-y binding
T2	Every-basis inversion and message invariance
T3	Complete-ring necessity versus smaller witnesses
T4	Quantitative attacker-to-complete-ring extraction
T5	Syntax, history, controls, sources, and prior art

Candidate A: RCF-Bind

RCF-Bind proposes a characteristic fiber indexed by the reduced trace and norm (t, n) of a primitive noncentral endomorphism α . The accepted pair contains α and its dual and satisfies

```
alpha + alpha_dual = [t]
alpha * alpha_dual = [n]
Delta = t^2 - 4n < 0.
```

The intended public equivalence identifies all accepted maps in one fixed trace/norm fiber through rational conjugacy in the definite quaternion algebra. This differs materially from endpoint and finite-evaluation relations. The producer correctly stopped with a typed obstruction rather than claiming a construction. [A]

The T1 owner independently rederived the conditional conjugacy argument, but T1 **breaks** because the packet supplies no public `SampleNoncentralEnd(E, D; rho)`, exact law, runtime, or success bound. Scalars and immediate isogeny/dual backtracks have $\Delta=0$ and are rejected, so they do not fill this first missing arrow. [VA]

T2 is **inconclusive**. Trace and norm are invariant under a valid unimodular basis change, and a packing argument bounds the number of norm-bounded lattice elements by $O(D^2)$. But no uniform inverse is specified with basis-input and map-conversion costs, full domain checks, retry guarantees, or a recovery function constant on every accepted conjugacy fiber. Conditional conjugacy does not make arbitrary encoding- or point-action-based message recovery invariant. [VA]

T3 and T4 **break**. One accepted α gives an evaluable embedding of a quadratic order; all polynomials in α and its dual remain in a two-dimensional rational algebra. That is strictly less output than a saturated rank-four basis. The packet has neither bare-E challenge sampling nor a theorem turning accepted attacker outputs into noncommuting rank growth, index certification, saturation, and evaluable full-ring generators. No uniform extraction probability, loss, runtime, or abort bound is supplied. [RA]

T5 **breaks** because the exact field/extension and rational-map byte grammar, unique function-field normal form, failure precedence, oracle history, and primary-source novelty boundary are incomplete. The review also corrected categorical absence language: a missing algorithm is “not supplied” or “not established,” not proved nonexistent. [T5A]

Candidate B: BDEB

BDEB proposes binding a degree-at-most-D endomorphism by its full action on an ordered list of $M=4D+1$ public points. Its honest support chooses scalar maps $[a]$; the transcript contains the points and images, and inversion enumerates the bounded scalar family. At the reviewed semantic level, the degree parallelogram identity and kernel-size bound can force two genuine degree-bounded maps agreeing on all listed points to be equal. [B] [VB]

T1 is nevertheless **inconclusive**. The exact public point-draw law, positive parameter regime, capped success probability, root selection, runtime, and common retry accounting are not fully fixed. Semantic map equality also does not substitute for a canonical byte representation. [VB]

T2 **breaks**. The stated inverse first validates the supplied complete basis, so its literal runtime cannot be invariant under arbitrary $GL_4(\mathbb{Z})$ shears whose encodings grow without bound. More importantly for the construction, the successful public enumeration ignores the basis entirely. [VB]

T3 and T4 **break** decisively on the honest support. The scalar preimage is publicly recoverable; arbitrarily many recovered witnesses remain on the rank-one scalar line and supply no noncentral generator, rank-four order, index, saturation, or evaluable complete basis. The simulator already knows the scalars it sampled. Fresh coins and repeated

success cannot turn this rank-one certificate into full EndRing output, and no generic IND-distinguisher to map-preimage compiler is supplied. [RB]

T5 breaks because exact map/mask serialization, quotient-function-field normalization, algorithm-specific malformed-input behavior, attempted-call rather than accepted-call accounting, coin visibility, and primary-source overlap remain incomplete. These specification gaps are additional to, not a repair for, the scalar-support collapse. [T5B]

Evaluation matrix and revision decision

Candidate	T1	T2	T3	T4	T5	Disposition
RCF-Bind	breaks	inconclusive	breaks	breaks	breaks	typed obstruction
BDEB	inconclusive	breaks	breaks	breaks	breaks	typed obstruction

The raw independence checker passed separately on the three owner reports for each candidate: every joint was owned and attested, blindness was respected, and controls were declared. Before any review returned, the coordinator added candidate-specific plans containing checker-required **blind_rederivation** quantity and **blind_from** fields. **PD-PLAN-SPLIT** is disclosed as a checker- shape-only correction; candidate bytes, owners, attacks, and evidence scope did not change. [Initial]

No revision was dispatched. Repairing only RCF-Bind’s sampler would leave its T2 unresolved and T3–T5 broken. Replacing BDEB’s scalar support would replace the mechanism that makes its public sampling and inversion work, forcing new **Sample**, **Inv**, extraction, and view definitions. The apparent shared gap— public noncentral generation outside an enumerable proper witness **and** preimage-to-complete-ring reconstruction—contains two independent missing algorithms. It is a new architecture, not one bounded repair. [Gap] [Decision]

Controls and inherited correction scope

The analytical reviews exercised leaked-witness/public-inverse, scalar and backtrack, finite-action/proper-witness, oriented/torsion-aided nearby-object, valid basis-shear, repeated-output rank, and full-rank-but-nonsaturated-order controls. These show where the submitted reasoning fails; they do not prove unconditional separations between OneEnd-like and EndRing problems. [RA] [RB]

The batch preserved the authorized J4/J5 correction semantics as inherited regression controls. The additive mapping remains printed pages 69–70 to PDF 74–75, printed 74 to PDF 79, and printed 75/Lemma 6.12 to PDF 80 for the previously recorded source bytes. No fresh primary PDF read occurred here and historical J4/J5 verdicts were not recoded. Likewise, normalized-**x**(**X**) claims retain their fixed-model, simultaneous-normalization, **p>3**, and exact-order **ord**(**P**)=**N>4D** scope; weakened order premises are controls, not valid-input attacks. No digit-loop output is inferred without complete tie-handling pseudocode. [Correction] [Method]

Requirement-to-evidence table

Requirement	Outcome	Evidence
Two materially distinct blind proposals	complete	[A], [B]
Immutable producer snapshot	complete	[Snapshot]
Independent T1–T5 owner review	complete	[VA], [VB], [RA], [RB], [T5A], [T5B]
Raw candidate-specific independence checks	pass twice	[Initial]
Compose without voting	complete	[Initial]
At most one decision-changing revision	skipped by frozen early-stop rule	[Gap], [Decision]

Requirement	Outcome	Evidence
Exact final counts and claim boundary	complete	[Final]
Primary-source novelty review	not performed	[T5A], [T5B]
Secure EndRing PKE or surviving interface	not established	[Final]

Exactly one next action

Study **ENDRING-PUBLIC-NONCENTRAL-SAMPLER-FEASIBILITY** before another PKE wrapper. On the exact bare-E EndRing distribution, either supply a polynomial public sampler for a canonically encoded noncentral evaluable endomorphism—with an exact law, inverse-polynomial success, runtime, retry bound, and controls— or retain a scoped obstruction at its first missing arrow. The output must not collapse to scalars, immediate backtracks, a leaked orientation/ideal/tag, or another efficiently enumerable proper witness family. [Final]

This action deliberately excludes the separate preimage-to-complete-ring reconstruction problem. It does not launch automatically and, even if solved, would not itself establish a secure PKE or EndRing reduction. [Final]

Source index

- [Method] [Frozen methodology](#)
- [Schema] [Binding-interface schema](#)
- [Plan] [Initial review plan](#)
- [A] [RCF-Bind packet](#)
- [B] [BDEB packet](#)
- [Snapshot] [Immutable producer snapshot](#)
- [VA] [Candidate A T1–T2 review](#)
- [VB] [Candidate B T1–T2 review](#)
- [RA] [Candidate A T3–T4 review](#)
- [RB] [Candidate B T3–T4 review](#)
- [T5A] [Candidate A T5 review](#)
- [T5B] [Candidate B T5 review](#)
- [Initial] [Initial composition](#)
- [Gap] [No-revision gap packet](#)
- [Decision] [Interim no-revision decision](#)
- [Correction] [Inherited J4/J5 correction addendum](#)
- [Final] [Final composition](#)